

Bespin Global Interview Assignment

Python AI Security Analyst Assistant

Goal

You are a Security Engineer on a Purple Team that needs to automate mundane tasks. Your goal is to build a small Python-based tool that can ingest, analyze, and correlate one or more security artifacts—such as extract from SIEM logs or a vulnerability scan report.

The challenge is to move beyond simple, known-signature matching and use a Generative AI model to analyze and contextualize unknown, anomalous, and potentially malicious activity. Your tool should act as an "AI-assisted analyst," helping to connect the dots between potential weaknesses and active threats. You MAY find real artifacts online from a CTF, your home lab, or other source OR you MAY generate synthetic data using LLMs, but you MUST explain how and why you generated the artifact and the strengths and weaknesses of your dataset.

Core Requirements

Your solution must be a Python script that accomplishes the following:

1. Ingest and store the artifact for analysis
2. Create one or more prompts that instruct the LLM on how to analyze the artifact.
3. Query an API based model to perform the analysis. You may use a self-hosted mode or API based model from Google, Microsoft, Anthropic, or OpenAI to analyze the artifact.
4. Generate a report: The script's final output must be a single, human-readable report printed to the console. This report should clearly summarize any threats, vulnerabilities, or system posture. Identify next steps for a human analyst.

Assignment Deliverables

1. **Python Script(s):** All Python code used to solve the challenge. You may use whatever libraries or SDKs you feel are appropriate but the code should be yours. You may use AI coding tools, but you should be able to explain which tools you used and why.
2. **requirements.txt:** A file listing all necessary Python libraries used. Use a tool such as uv or Poetry to manage virtual environments.
3. **README:** A brief markdown file containing clear instructions on how to set up the environment and run your script (including how to provide an API key) and include a list of known weaknesses or constraints with your tool, and future enhancements you would make if you had more time.

Time Assumptions

If you are an experienced Python developer with experience using AI to generate code AND have experience with LLM APIs and/or agent frameworks, this should take you no more than 1-2 hours. However, if you are not familiar with querying models, it may take you longer to complete this assignment. We do not expect this to be a production-grade tool or be full-featured. This goal assesses your development skills and ability to rapidly define and implement a solution using AI.

Evaluation Criteria

- **Code Quality:** Is the code clean, efficient, and well-structured?
- **Security Acumen:** Does your solution show a strong "purple team" mindset in terms of understanding attacks, defense
- **GenAI Skill:** How effective was your prompting strategy? Did you successfully provide context to the LLM? How well did your script handle the API interaction and its response?
- **Final Report:** Is the output clear, concise, and actionable for a security team?

Next Steps

During a follow-on interview you will meet with Bespin Engineers and demo your tool and talk through your design and analysis process. If you have any questions feel free to email matt.franz@bespinglobal.com. Once you are done, let the recruiter team know so they can schedule this session.